

Axioms of Probability

1. Non-negativity: For any event A , $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$. This can be extended to $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for mutually exclusive events $A_1, A_2, \dots, A_n$.

Basic Properties of Probability:

$$\begin{aligned}\forall A \in S, P(A) &< 1 \\ P(A^c) &= 1 - P(A) \\ P(\emptyset) &= 0 \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ A \subseteq B &\implies P(A) \leq P(B)\end{aligned}$$

Counting:

Permutation — An arrangement of objects in a specific order. $P(n, r) = \frac{n!}{(n-r)!}$

Combination — A selection of objects without regard to the order. $C(n, r) = \frac{n!}{r!(n-r)!}$

Binomial Coefficient — The number of ways to choose k successes in n independent Bernoulli trials. $\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$

Number of Subsets — The number of subsets of a set with n elements is 2^n .

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication Rule: — $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

$$P(\bigcap_{i=1}^n A_i) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_n|\bigcap_{i=1}^{n-1} A_i)$$

Law of Total Probability: — If $B_1, B_2, \dots, B_n$ are mutually exclusive and exhaustive events, then for any event A : $P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$

Bayes' Theorem: —

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

Independence:

Two events A and B are independent if and only if $P(A|B) = P(A)$ and $P(A \cap B) = P(A)P(B)$.

If A and B are independent, then B and A , A and B^c , A^c and B , A^c and B^c are also independent.

Random Variables An RV is a function that assigns a real number to each outcome in the sample space of a random experiment.

Discrete RV — A random variable that can take on a countable number of distinct values.

Continuous RV — A random variable that can take on any value within a given range or interval.

$X: S \rightarrow \mathbb{R}$ Where S is the event space and X is the random variable.

PMF — $f(x) = P(X = x)$ (discrete)

$$P(a \leq X \leq b) = \int_a^b f(x)dx \text{ (continuous)}$$

CDF — $F(x) = P(X \leq x) = \sum_{y \leq x} P(X = y)$ (discrete)

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt \text{ (continuous)}$$

$$P(a \leq X \leq b) = F(b) - F(a)$$

Bernoulli Random Variable — A discrete RV that takes the value 1 with probability p and the value 0 with probability $1 - p$.

Expected Value — $E(X) = \mu_X = \sum_x xP(X = x)$ or $= \int_{-\infty}^{\infty} xf(x)dx$

$$E(h(X)) = \sum_x h(x)P(X = x) \text{ or } E(h(X)) = \int_{-\infty}^{\infty} h(x)f(x)dx$$

$$E(X + Y) = E(X) + E(Y)$$

$$E(X^2) = \sum_x x^2 p(x) \text{ or } = \int x^2 f(x)dx$$

Variance — $\sigma_X^2 = E[(X - \mu_X)^2] = E(X^2) - (E(X))^2$ or $\sigma_X^2 = \int (x - \mu_X)^2 f(x)dx$

Standard Deviation — $\sigma_X = \sqrt{Var(X)}$

Continuous Distributions:

Normal — $X \sim \mathcal{N}(\mu, \sigma^2)$

PDF: — $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ for $-\infty < x < \infty$ else 0

Standard Normal — $Z \sim \mathcal{N}(0, 1)$ $Z = \frac{X-\mu}{\sigma}$

$P(a \leq X \leq b) = P(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$
 z_α is the z-score such that $P(Z \geq z_\alpha) = \alpha$ or the $(1 - \alpha) \cdot 100$ th percentile

Exponential — $X \sim \text{Exp}(\lambda)$

PDF: — $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ else 0

CDF: — $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ else 0

Mean: — $E(X) = \frac{1}{\lambda}$

Variance: — $Var(X) = \frac{1}{\lambda^2}$

Uniform — $X \sim U(a, b)$

PDF: — $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$ else 0

CDF: — $F(x) = \frac{x-a}{b-a}$ for $a \leq x \leq b$, 0 for $x < a$, 1 for $x > b$

Survival Function: — $S(x) = 1 - F(x) = \frac{b-x}{b-a}$ for $a \leq x \leq b$, 1 for $x < a$, 0 for $x > b$

Mean: — $E(X) = \frac{a+b}{2}$

Variance: — $\sigma^2 = \frac{(b-a)^2}{12}$

Jointly Distributed Random Variables:

$$P(x, y) = P(X = x, Y = y)$$

$$\text{Trick: } P(X = Y) + P(Y > X) + P(Y < X) = 1$$

Marginal PMF — $P_X(x) = \sum_y P(x, y)$ and $P_Y(y) = \sum_x P(x, y)$

X and Y are independent $\iff P(x, y) = P_X(x)P_Y(y)$

Conditional PMF — $P_{Y|X}(y|x) = \frac{P(x, y)}{P_X(x)}$

Expected Value — $E(h(X, Y)) = \sum_x \sum_y h(x, y)P(x, y)$

$$E(h(X, Y)) = \iint h(x, y)f(x, y)dxdy$$

Covariance — $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y)$

$Cov(X, Y) = 0$ if X and Y are independent.

Correlation Coefficient — $\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$

$$-1 \leq \rho_{XY} \leq 1$$

Properties $Cov(X, X) = Var(X)$ $Cov(X, Y) = Cov(Y, X)$

$$Cov(aX, bY) = abCov(X, Y) \quad Cov(X+Y, Z) = Cov(X, Z) + Cov(Y, Z) \quad \exists a, b \in \mathbb{R} \ni Y = aX + b$$

Random Samples and Statistics:

A random sample of size n is a set of n independent and identically distributed (i.i.d.) random variables $X_1, X_2, \dots, X_n$ drawn from a population with a common distribution.

Sample Mean — $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Sample Variance — $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

$$E(\bar{X}) = \mu$$

$$Var(\bar{X}) = \frac{\sigma^2}{n}$$

When sampling from a normal population, $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$

Linear Combinations — $Y = a_1X_1 + a_2X_2 + \dots + a_nX_n$

$$E(Y) = a_1E(X_1) + a_2E(X_2) + \dots + a_nE(X_n)$$

$Var(Y) = a_1^2Var(X_1) + a_2^2Var(X_2) + \dots + a_n^2Var(X_n)$ if X_i are independent.

If $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $Y \sim \mathcal{N}(a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n, a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \dots + a_n^2\sigma_n^2)$